

KPI Catalogue

Analytics Knowledge Base — TPC-H Data Platform

Catalog: samples | Schema: tpch | Databricks

This catalogue serves as the single source of truth for all business KPIs surfaced through the enterprise analytics platform. Each entry describes how the metric is derived, which tables and columns feed into it, and how stakeholders should interpret the output. Every metric is computed against the TPC-H dataset available on Databricks (catalog: samples, schema: tpch).

1. Revenue Metrics

1.1 Gross Revenue

Gross Revenue represents the total monetary value collected from customers across all shipped line items within a specified time window. Discounts are factored in at the line-item level before aggregation.

Computation: $\text{SUM}(l_extendedprice \times (1 - l_discount))$

Tables involved: samples.tpch.lineitem, samples.tpch.orders

Key columns:

- `l_extendedprice` — the base unit price for each line item before discounting
- `l_discount` — proportional discount applied per line item (range: 0.00–0.10)
- `l_shipdate` — primary date filter for shipment-window analysis
- `o_orderdate` — alternate date filter when querying at the order level

Sample use case:

→ Break down total revenue by nation for the most recent fiscal year.

1.2 Average Order Value (AOV)

AOV measures the typical spend associated with a single customer order. It is obtained by dividing aggregated revenue across all shipped line items by the number of unique orders in the same period. A consistently rising AOV signals that customers are either purchasing higher-priced goods or consolidating more items per transaction — both indicators of improved sales quality.

Computation: $\text{SUM}(l_extendedprice \times (1 - l_discount)) \div \text{COUNT}(\text{DISTINCT } o_orderkey)$

Tables involved: samples.tpch.lineitem, samples.tpch.orders

Sample use case:

→ Compare AOV across market segments to prioritise high-value customer cohorts.

1.3 Revenue by Region / Nation

This metric disaggregates Gross Revenue along geographic dimensions — either at the regional or national level — enabling territory-level performance benchmarking and commercial planning.

Tables involved: lineitem, orders, customer, nation, region

Join path: lineitem → orders (o_orderkey) → customer (o_custkey) → nation (c_nationkey) → region (n_regionkey)

2. Order Metrics

2.1 Total Order Count

Tracks the volume of unique orders placed within a given reporting window. This is a foundational throughput indicator used across supply chain, sales, and operations teams.

Computation: COUNT(DISTINCT o_orderkey)

Tables involved: samples.tpch.orders

2.2 Order Fulfillment Rate

Quantifies what percentage of orders have been fully dispatched — all associated line items shipped. This metric is a direct proxy for supply chain reliability and operational throughput.

Computation: COUNT(fully shipped orders) ÷ COUNT(all orders) × 100

Tables involved: samples.tpch.orders, samples.tpch.lineitem

Order status codes:

- O — Order is open / pending
- F — Order is fully fulfilled (all items shipped)
- P — Order is partially shipped (some items still pending)

2.3 Average Shipping Lag

Measures the mean elapsed time (in days) from when an order is placed to when its line items are dispatched. Lower values indicate a more responsive supply chain; sustained increases may signal procurement or logistics bottlenecks.

Computation: AVG(DATEDIFF(l_shipdate, o_orderdate))

Tables involved: samples.tpch.lineitem, samples.tpch.orders

3. Supplier & Cost Metrics

3.1 Procurement Cost

Captures the aggregate spend on inventory by multiplying the per-unit supplier cost against the available stock quantity for every supplier–part combination. This metric supports procurement budgeting and vendor cost benchmarking.

Computation: $\text{SUM}(\text{ps_supplycost} \times \text{ps_availqty})$

Tables involved: samples.tpch.partsupp, samples.tpch.supplier

3.2 Supplier Performance Ranking

Ranks vendors by the total revenue generated through their fulfilled line items. High-revenue suppliers represent critical dependencies in the supply chain and are prioritised in vendor management reviews.

Computation: $\text{SUM}(\text{l_extendedprice} \times (1 - \text{l_discount}))$ grouped by s_name

Tables involved: samples.tpch.lineitem, samples.tpch.supplier

4. Quick Reference — All Metrics

The table below provides a consolidated view of all KPIs, their formulas, source tables, permitted roles, and query scope.

Metric	Computation Formula	Source Table	Access Role	Scope
Gross Revenue	$\text{SUM}(\text{price} \times (1 - \text{discount}))$	lineitem	Finance / Exec	Shipment-level
Avg Order Value	$\text{Gross Revenue} \div \text{DISTINCT order count}$	lineitem + orders	Analyst / Finance / Exec	Order-level
Total Orders	$\text{COUNT}(\text{DISTINCT order key})$	orders	Analyst / Exec	Order-level
Fulfillment Rate	$\text{Fully shipped orders} \div \text{All orders} \times 100$	orders + lineitem	Analyst / Exec	Order-level
Ship Lag (Avg)	$\text{AVG}(\text{ship date} - \text{order date})$	lineitem	Finance / Exec	Line item-level
Procurement Cost	$\text{SUM}(\text{unit cost} \times \text{stock quantity})$	partsupp	Finance / Exec	Supplier-level
Revenue by Region	Gross Revenue grouped by geography	Multi-table join	Exec	Regional

5. Date Filtering Conventions

The TPC-H dataset spans order dates from 1992 through 1998. When queries reference relative time expressions such as "last year", "this year", or "recent", interpret them against this dataset range — treating 1997–1998 as the most current period.

Filtering guidelines:

- Order-level queries → filter on o_orderdate
- Shipment-level queries → filter on l_shipdate
- "Last year" in relative queries → WHERE YEAR(o_orderdate) = 1997

Example:

```
SELECT nation, SUM(l_extendedprice * (1 - l_discount)) AS gross_revenue
FROM samples.tpch.lineitem l
JOIN samples.tpch.orders o ON l.l_orderkey = o.o_orderkey
WHERE YEAR(o.o_orderdate) = 1997
GROUP BY nation ORDER BY gross_revenue DESC;
```